

PRODUCT & BI ANALYST — DATA CHALLENGE

Growth runs on first purchases

What \$946K of trailing revenue says about customers, churn and a free-shipping policy — and three moves to act on it.

Monthly revenue has doubled in a year — and 84% of it comes from people buying for the first time

Revenue, Aug 2026 \$115K +111% vs Aug 2025	Revenue from first-time buyers, last 12 mo 84% returning customers: 12–21% a month	First-time buyers who return within a year 1 in 6 16.2%; those who do take a median 412 days	First order as share of 24-month value 89% \$85 of \$95
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WHAT THE DATA SAYS

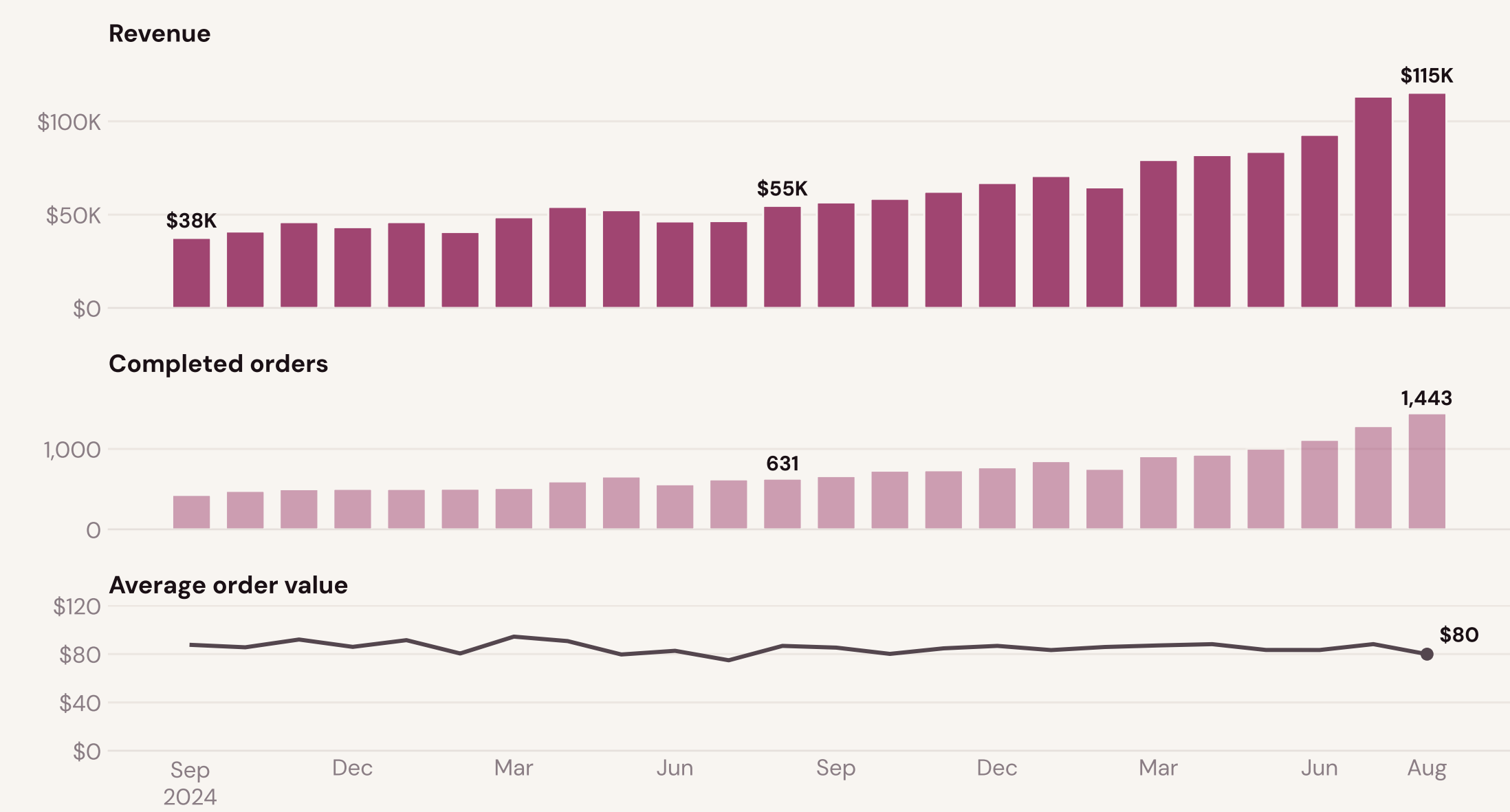
- 1 **Growth is volume, not basket size.** Orders +129% year over year; average order value did not grow (\$87 → \$80).
- 2 **Repeat purchase is rare and slow, so growth depends on acquisition** — 70% of it through one channel, Search.
- 3 **Free shipping over \$100 would lose money.** Simulated, gross profit after shipping falls 26%: 91% of the parcels it pays for go to orders already over \$100.

WHAT I RECOMMEND

- 1 **Win the second purchase in the first 60 days**, when repeat peaks. Lifting 12-month repeat by 3 pts ≈ \$25K a year.
- 2 **Free shipping on the second order, not over \$100** — domestic first. Breaks even at +1.4 pts of 90-day repeat.
- 3 **Fix measurement before scaling:** agree what a “sale” is, move churn to 12 months, automate the QA checks.

Revenue grew 111% year over year on order volume — average order value did not grow

Trailing-12-month revenue is \$946K, up 70% on the prior 12 months. The typical month grows 4.5% on the one before.



+129%

completed orders, Aug 2026 vs a year earlier

\$75–\$94

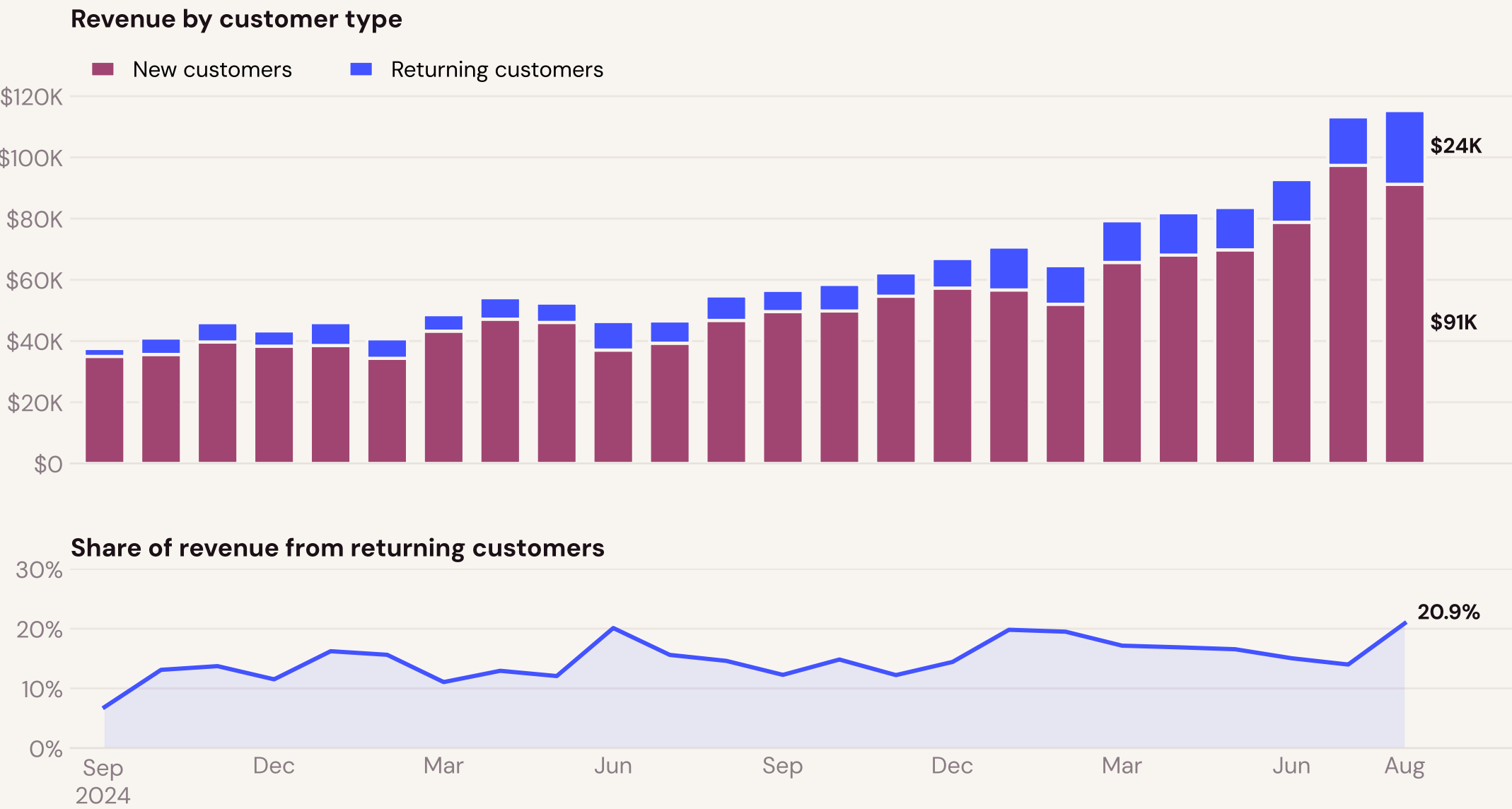
range of monthly average order value over two years

4.5%

median month-over-month revenue growth, last 12 months

About 8 in 10 revenue dollars every month come from first-time buyers

Returning customers are not the problem — they spend more per head (\$89 vs \$80 in Aug). There are just very few of them each month.



84%

of revenue from first-time buyers, last 12 months

-8%

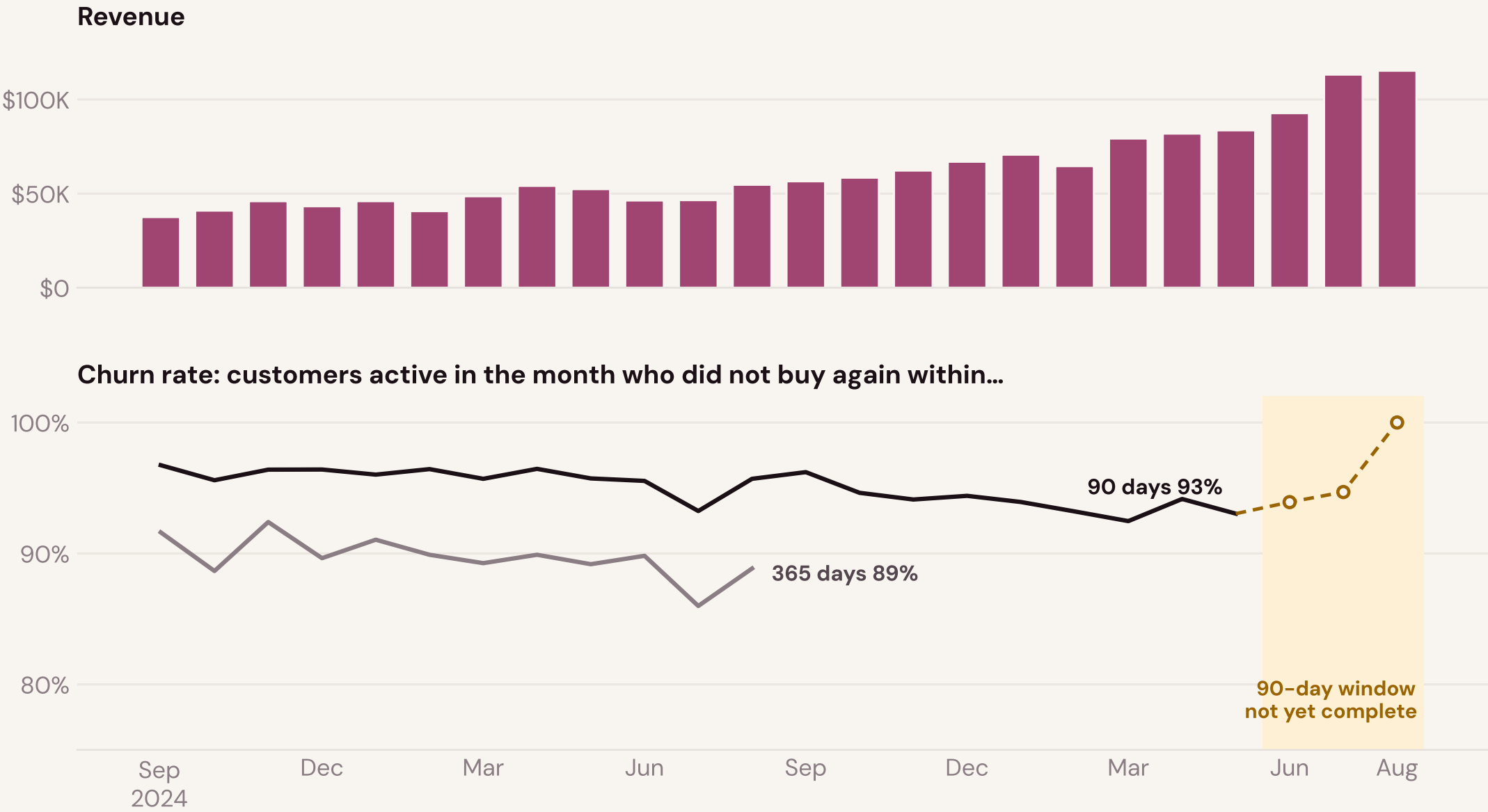
revenue if new customers fell 10% — there is little repeat base to absorb it

+12%

spend per returning customer vs per new customer

94% of active customers don't buy again within 90 days — and even over a full year, 90% don't

Revenue and churn rise together: the business grows by adding customers, not by keeping them. A longer window lowers churn only slightly — repeat is rare, not just slow.



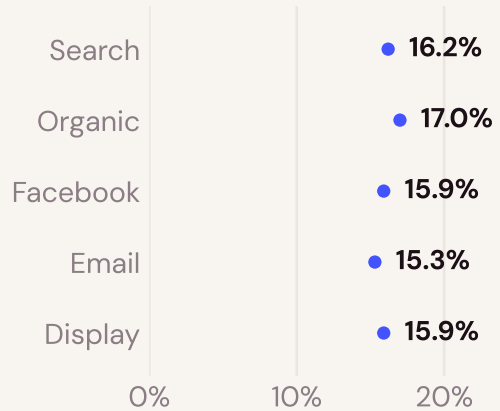
95.8% → 89.7%

same months (Sep 2024 – Aug 2025),
90-day vs 365-day window

94.1% → 88.5%

if a "purchase" includes shipped and
processing orders, not only 'Complete'

12-month repeat rate by channel —
identical everywhere



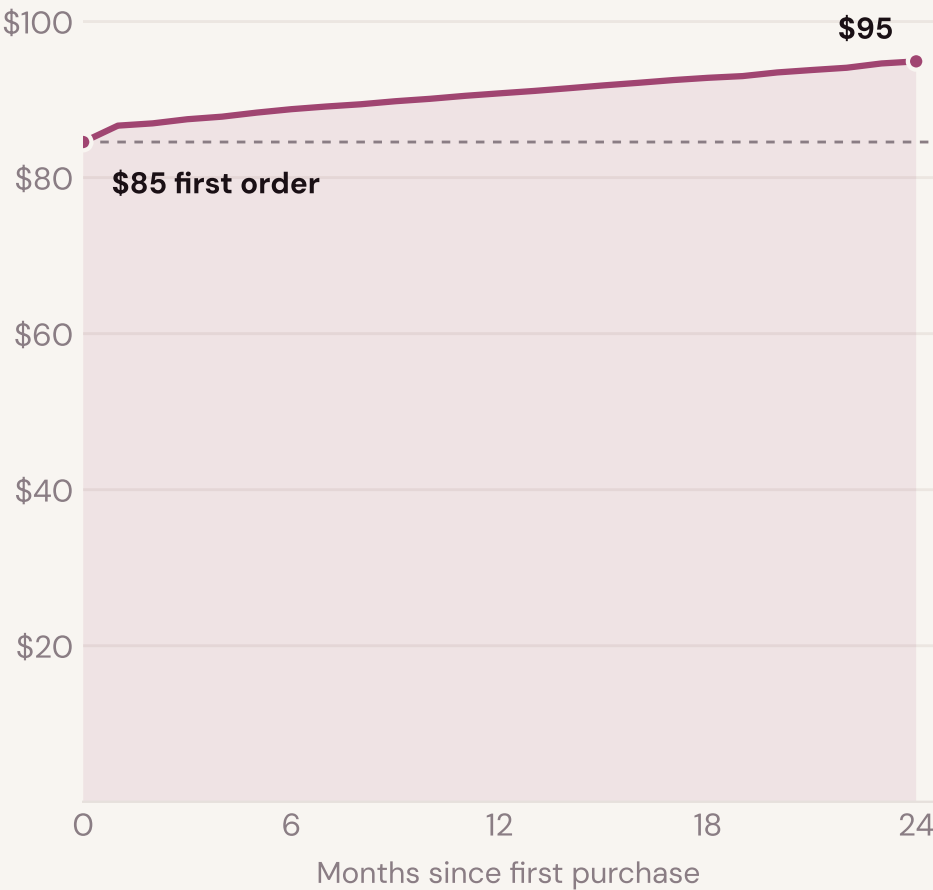
Only 1 in 6 first-time buyers returns within a year — so the first order is 89% of two-year value

Month 1 is the best chance of a second purchase (median 3.1% of a cohort, vs 0.9% in months 2–8). Those who come back later take a median of 412 days.

Cohort retention — % of each first-purchase cohort buying again in month N

	M1	M2	M3	M4	M5	M6	M7	M8
Jan 2025 (430)	2.8	0.5	0.9	0.2	0.7	0.7	0.2	0.2
Feb 2025 (430)	1.4	0.9	1.2	0.9	0.2	1.2	0.5	1.2
Mar 2025 (446)	3.1	0.5	0.9	1.4	1.6	0.2	0.7	0.2
Apr 2025 (504)	2.2	1.0	0.6	0.4	1.2	0.8	1.0	0.8
May 2025 (572)	3.3	1.1	0.3	1.4	1.2	0.3	0.2	1.4
Jun 2025 (469)	3.0	0.8	0.8	0.2	0.2	0.8	0.4	0.6
Jul 2025 (521)	4.6	1.7	1.1	0.6	0.8	1.1	0.8	1.1
Aug 2025 (537)	2.6	1.1	0.9	0.6	0.7	1.1	0.7	0.7
Sep 2025 (566)	2.6	0.5	0.9	1.1	0.5	0.7	0.5	1.1
Oct 2025 (613)	2.9	1.0	1.6	1.6	2.0	1.1	1.3	1.1
Nov 2025 (633)	3.6	0.6	2.5	1.1	0.6	0.5	1.7	1.3
Dec 2025 (659)	3.2	1.4	1.1	1.5	1.2	1.8	0.9	2.1
Jan 2026 (686)	3.6	1.3	1.3	2.0	0.9	1.8	1.5	
Feb 2026 (609)	4.9	1.8	0.8	1.8	0.7	2.0		

Average cumulative spend per customer — 10,846 customers with 24 months of history



Three moves: win the second purchase early, point free shipping at it, and fix measurement first

1

Win the second purchase in the first 60 days

Month 1 is when repeat peaks (3.1% of a cohort vs 0.9% later), and returning customers already spend 12% more per head.

- TARGET

Every first-time buyer, by **moment** not channel — channels behave identically.
- PRIMARY

12-month repeat rate vs a 10–20% holdout (today 16.2%).
- SECONDARY

Days to 2nd purchase (412); share of revenue from returning (16%).
- GUARDRAIL

Incentive cost per incremental order; unsubscribe rate.

A +3-pt lift (16.2% → 19.2%) ≈ **\$25K** a year, 2.6% of revenue — counting only the first repeat order.

2

Free shipping on the *second* order, not over \$100

Over-\$100 free shipping pays for parcels customers were already buying: 91% of them. Only 4.4% of buyers repeat within 90 days, so a second-order offer pays mostly for new behaviour.

- TARGET

US first-time buyers first (23% of revenue). Abroad, shipping eats 78% of an order's profit — use a non-shipping incentive.
- PRIMARY

90-day repeat rate vs holdout (today 4.4%).
- GUARDRAIL

Gross profit after shipping.

Breaks even at **+1.4 pts** of 90-day repeat in the US; would need +15.5 abroad.

3

Fix measurement before scaling decisions

'Complete' samples 25% of orders at random, a 90-day churn window is too short for this repeat cycle, and the session funnel cannot measure conversion.

- DO

Agree the revenue-recognition point with finance; report churn on 12 months; ship the QA checks as scheduled tests; publish curated views for self-service and text-to-SQL.
- MEASURE

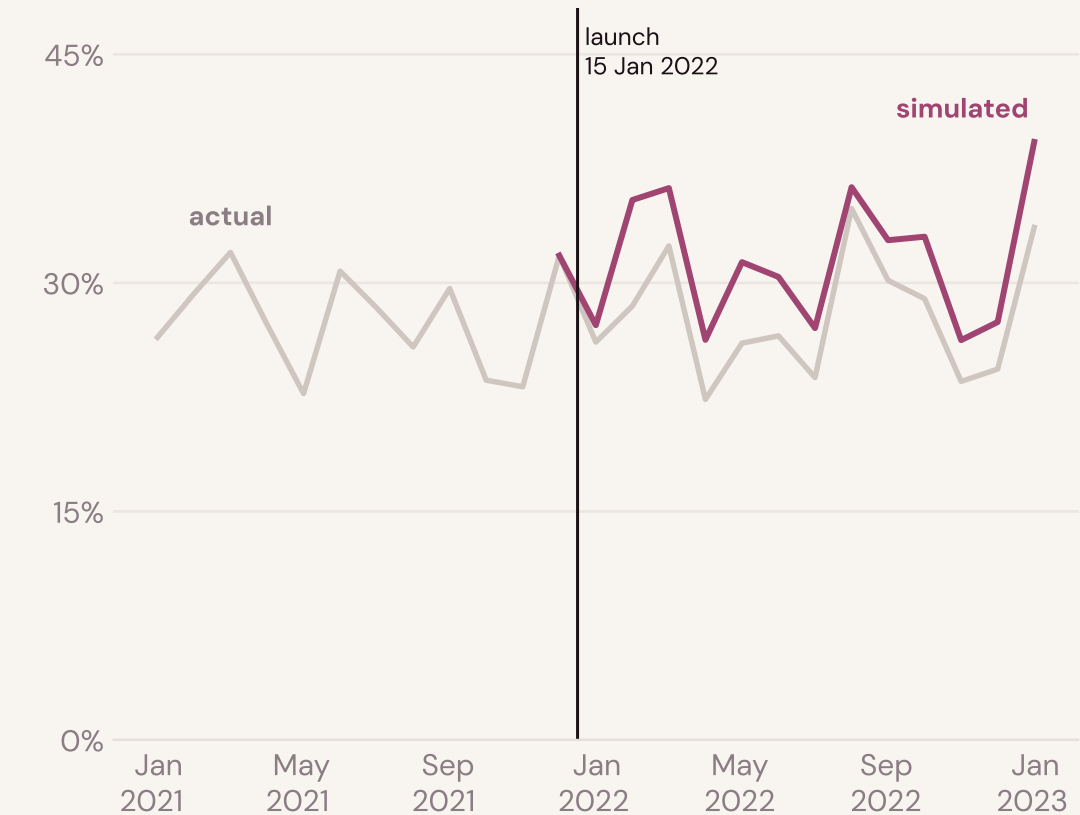
QA pass rate, data freshness, share of KPIs on governed definitions.

Makes 1 and 2 **measurable** in the first place.

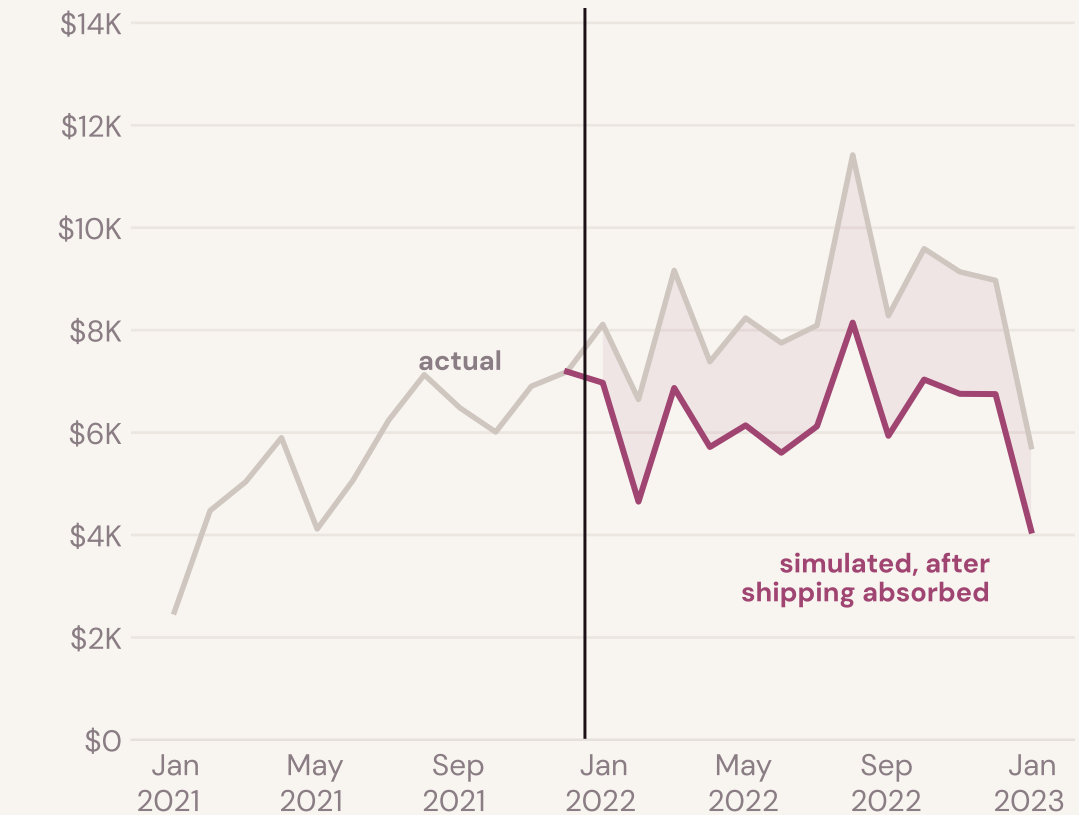
Simulated, free shipping over \$100 moves orders over the line — and cuts gross profit after shipping by 26%

The policy is hypothetical, so its effect is injected: 30% of \$70–\$99 orders top up after launch. Orders over \$100 rise from 27.8% to 31.5% and revenue +1.1% — but the business now pays for parcels on every eligible order.

Share of orders over \$100 — what the policy is meant to move



Gross profit per month — after the shipping the policy absorbs



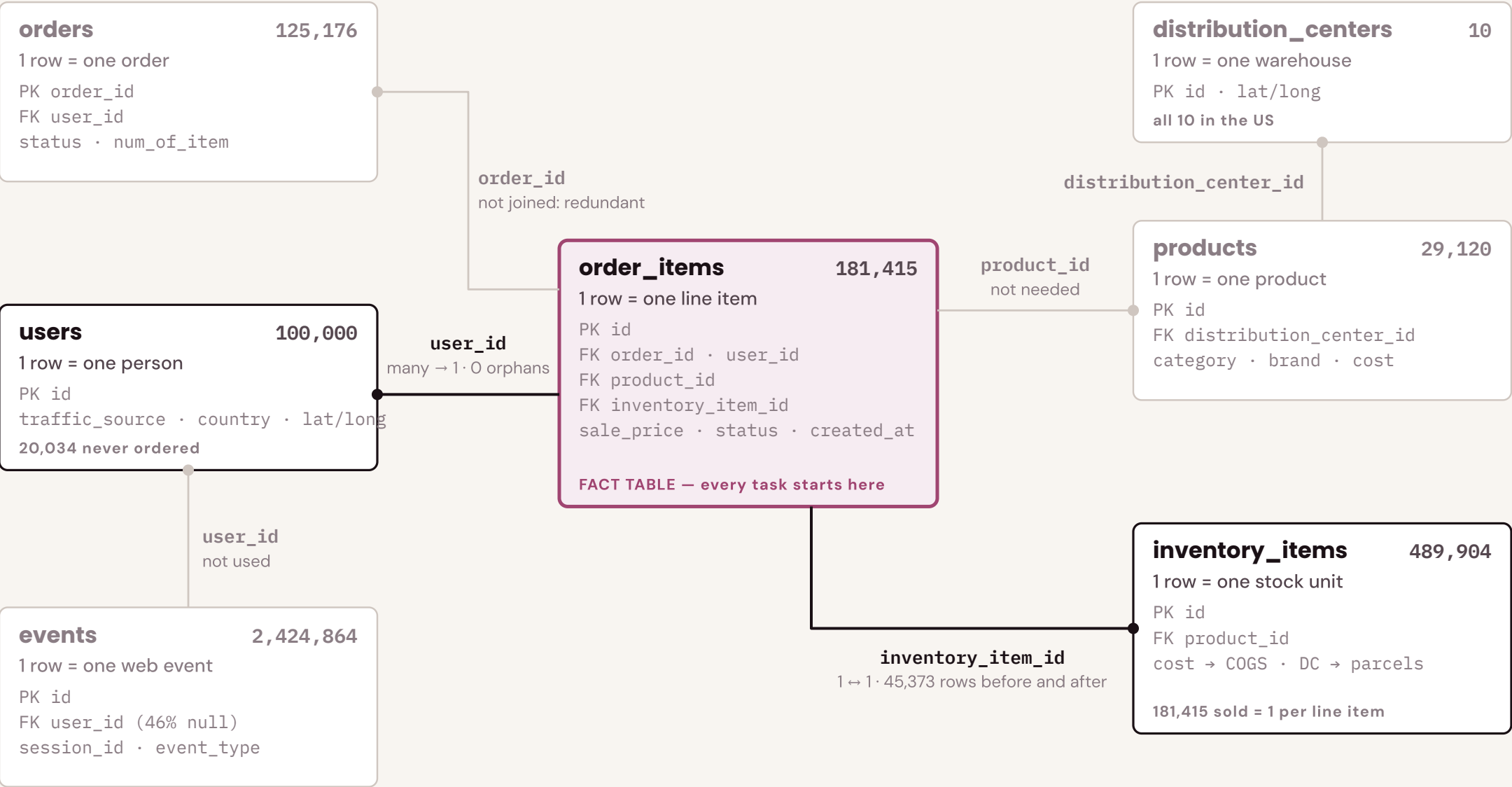
< \$1

what a parcel may cost before the policy loses money (\$0.95 US, \$0.82 abroad) — 91% of parcels go to orders already over \$100

How I would test it for real: a randomized holdout. Pre/post cannot separate it from growth, and with ~500 orders a quarter only effects ≥8 pts are detectable.

Data needed: shipping fees and parcel costs, a treatment flag, cart-abandonment events, returns cost.

One fact table and two joins — each checked to add columns, never rows

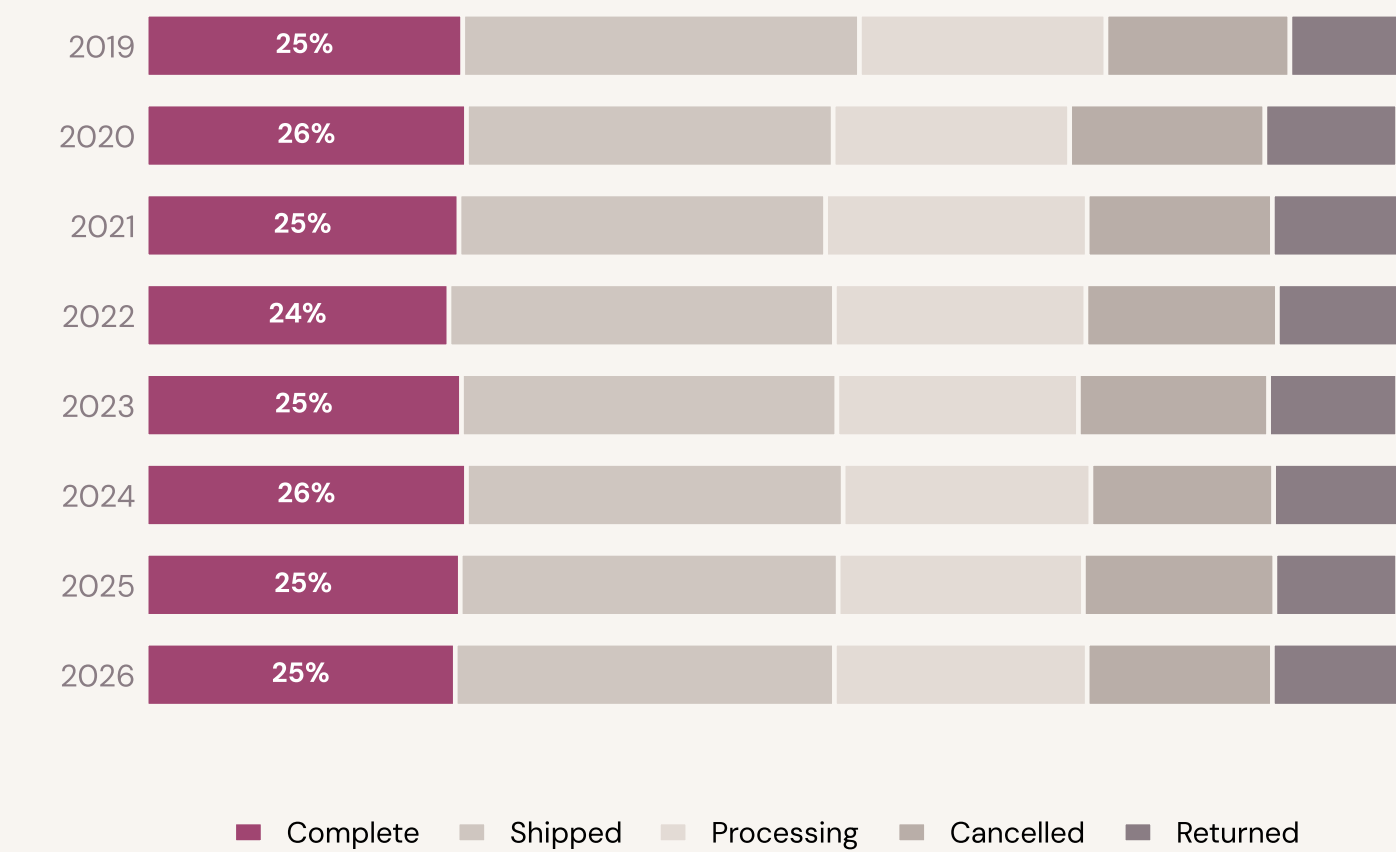


JOIN RULES APPLIED

1. **Pick the grain first.** `order_items` is one row per line item, so `orders` and `customers` are counted with `DISTINCT`.
2. **Count rows before and after every join.** `inventory_items`: 45,373 before, 45,373 after.
3. **Skip joins that add nothing.** `order_items` already carries the order's `user_id` and `status` (0 mismatches), so `orders` is never joined.
4. **Aggregate before joining one-to-many.** Task D rolls line items up to one row per order before attaching users.

Four things about this data change the numbers, and none of them is in the brief

Order status mix by year of order — a 2019 order is still 'Shipped' today



632 rows dated after today (2026-09-24). The count shrinks daily as today catches up — proof they are ahead of time. Every window caps at today.

25% of line items are 'Complete', the same share every year — status is random, so the brief's definition samples a quarter of real orders.

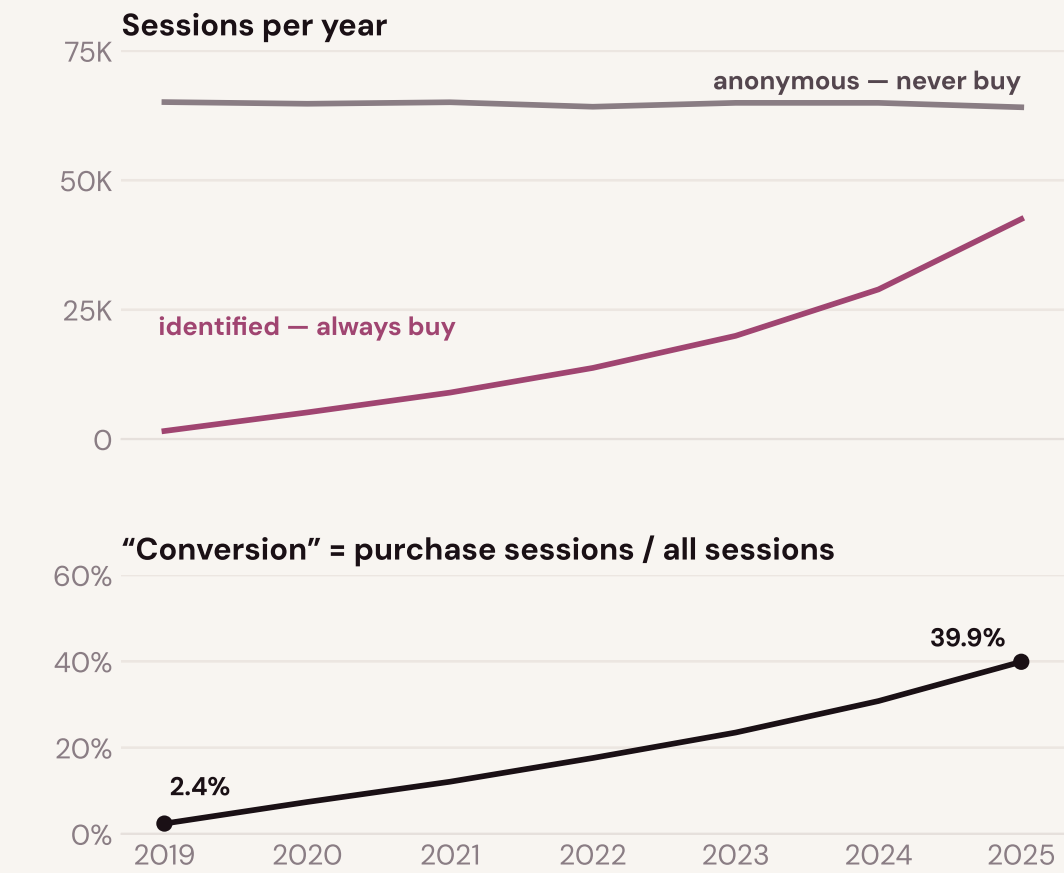
5.6 pts of churn come from that sampling alone (94.1% vs 88.5%).

0 columns empty, and every null is structural — e.g. shipped_at is null for exactly the Processing + Cancelled items. events.user_id is 46% null (anonymous sessions); IPs are random per session, so it cannot be backfilled — and no Part 1 metric uses it.

The session funnel can't measure conversion: anonymous traffic is a constant, so "conversion" climbs from 2% to 40% on its own

Every identified session ends in a purchase; no anonymous session ever does. Underneath, theLook places each order at a random date between signup and today — which is also why every calendar trend here rises into the present.

events, by year — fixed paths, fixed anonymous volume



Order timing test — 42,303 single-order customers



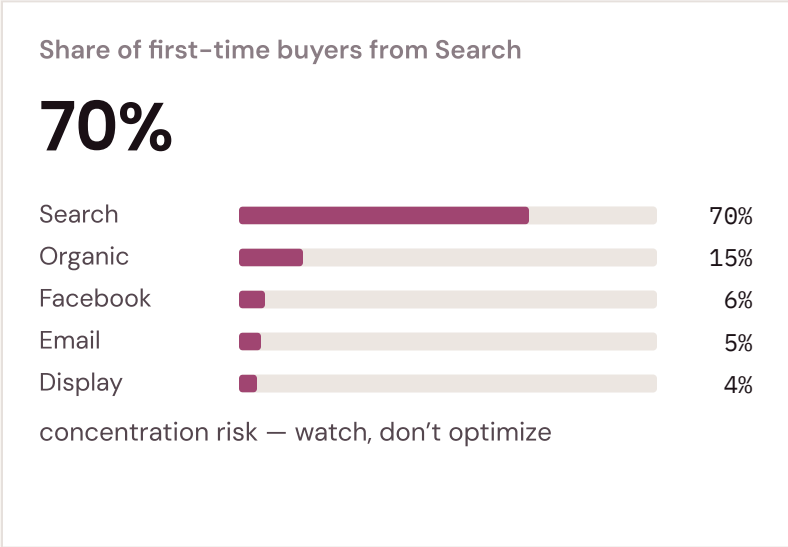
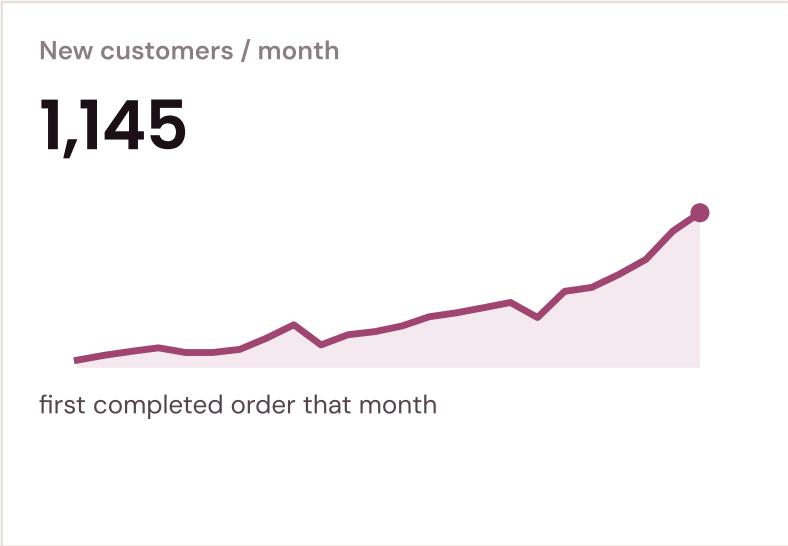
The funnel this data can support — user level, accounts ≥ 1 year old

- 100 signed up
- 80 placed an order
- 27 activated — first completed order
- 12 placed a second order

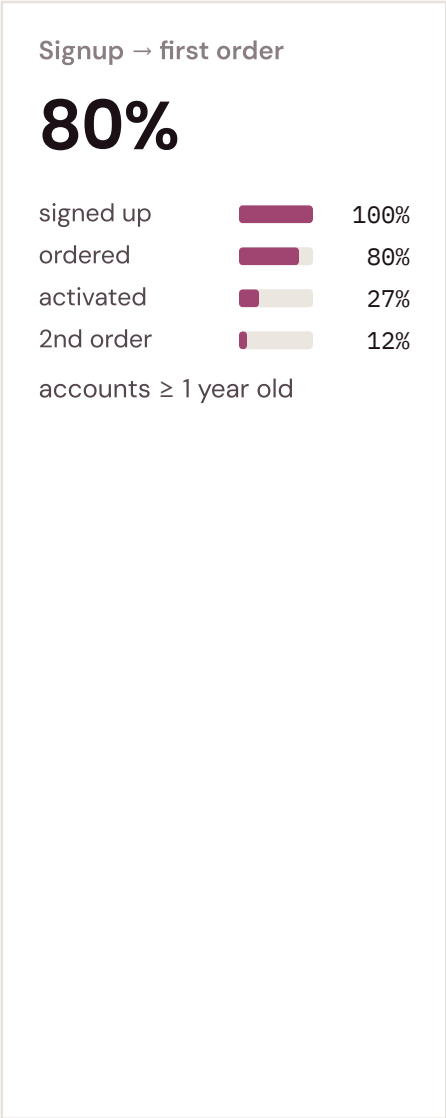
±0.6 pts
spread across the five acquisition channels. No channel converts or repeats differently.

Seven metrics, one per question leadership asks — with the data-health check that makes them trustworthy

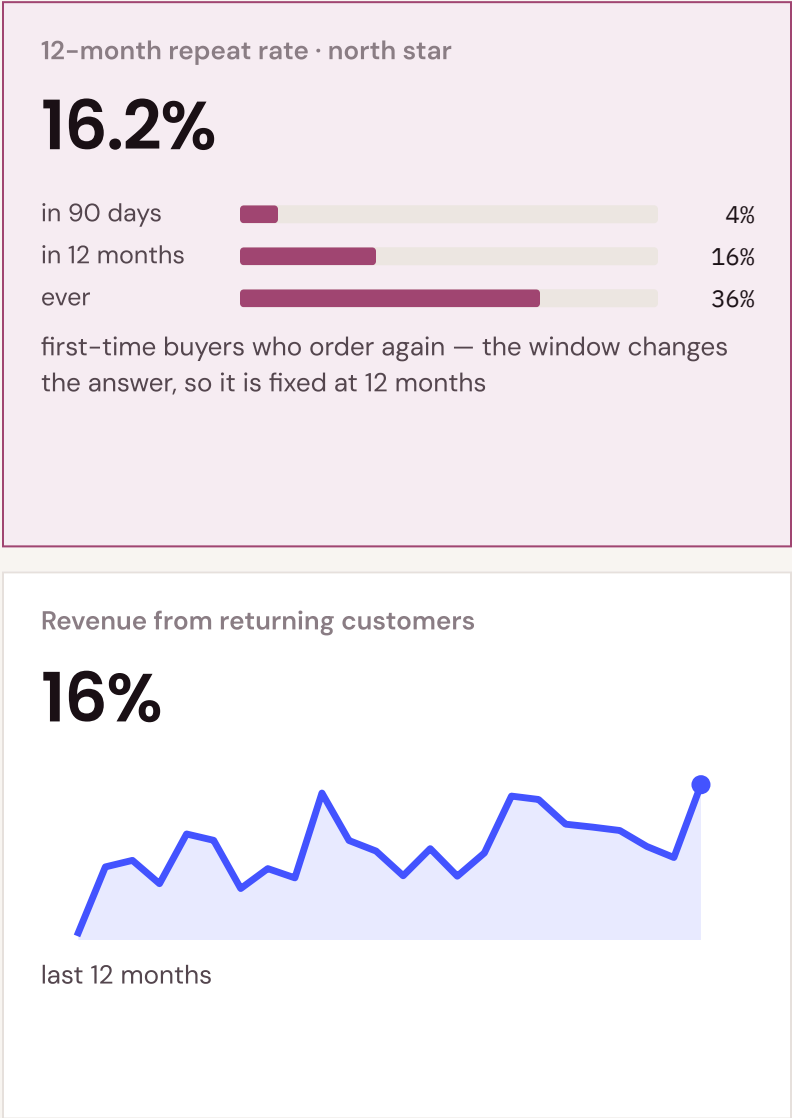
ACQUISITION



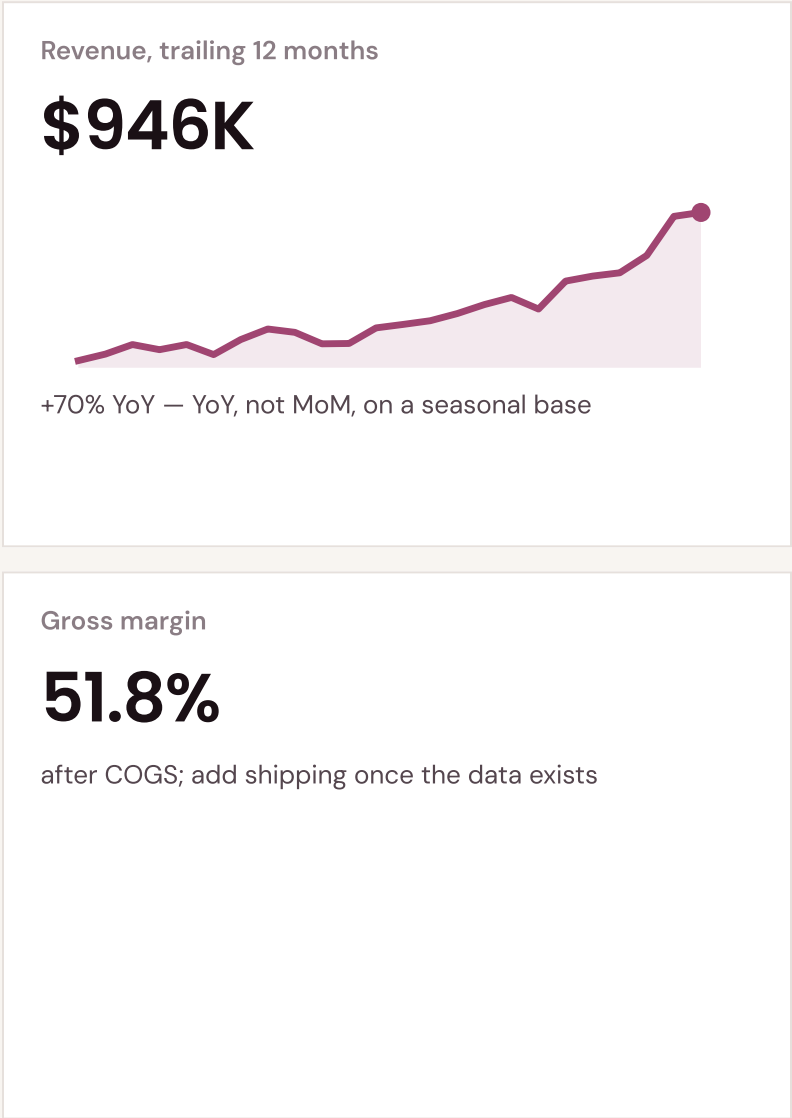
ACTIVATION



RETENTION



MONETIZATION



DATA HEALTH

✓ last complete month: Aug 2026

✓ future-dated rows excluded: 632

✓ QA checks passing: 4 / 4

✓ as of 2026-09-24 UTC

AI wrote the first draft of every query; a check that could fail decided whether to keep it

WHERE IT SPED THINGS UP • Profiling the seven tables, every null and every join key in minutes • First drafts of CTE-structured SQL for all four tasks • Rewriting Task C with LEAD(): identical output, 77× less compute • Generating this deck reproducibly from the query outputs	WHERE CHECKS OVERRULED IT • Text extraction silently dropped one sentence of the brief — re-reading the source recovered it • A Task D column was 100% or 0% by construction — rebuilt at monthly grain • A 100% churn month — the edge of the data, not an event • A session funnel showing conversion rising 17× — a constant denominator • A funnel that was not nested — each stage now a strict subset of the one before
WHAT I WOULD AUTOMATE NEXT • Run the QA queries on a schedule and alert before a dashboard refreshes • An LLM-drafted weekly KPI narrative, with every figure pulled from SQL, never computed by the model • Text-to-SQL over curated, documented views rather than raw tables	CHECKS THAT SHIP WITH THE REPO • Row counts before and after every join • Identity assertion: new + returning = active, every month • Rewrites verified row-for-row against the original • Definition sensitivity: churn under two definitions and two windows

EXAMPLE PROMPT

“The last three months return churn near 100%. Before assuming that is real: what in the data boundary could produce it, and write the check that would tell an artifact from a genuine spike.”